

DevOps Engineer

Step 3: Learn Docker

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

Docker packages applications and their runtime dependencies into images that can be instantiated as containers. The key conceptual distinction is that containers are isolated processes sharing the host kernel, not complete virtual machines.

Learn images, layers, Dockerfiles, containers, volumes, networks, registries and Compose. Production-quality containerization also involves non-root execution, small/pinned base images, secret management, health checks and useful logs.

Core Concepts — Detailed Breakdown

- Image vs container: an image is a reusable template; a container is a running instance.
- Layers: Docker images are built from layered filesystem changes.
- Dockerfile: instructions define how an image is constructed.
- Volumes: persistent data should not depend on a container's writable layer.
- Networks: containers communicate through virtual networks and service names.
- Compose: define multi-container development stacks declaratively.
- Registry: store and distribute images.
- Security: avoid embedding secrets, minimize packages, scan images and use least privilege.
- Operations: logs, health checks, resource limits and graceful shutdown.

What You Should Be Able To Do

- Write a Dockerfile.
- Build and tag an image.
- Map ports.
- Use a volume.
- Create a Compose stack.
- Inspect logs and container metadata.

Hands-on Project / Practice

- Containerize a Flask/Node application with a database. Add a Compose file, health check, environment configuration and persistent database volume. Document how to rebuild from scratch.

Recommended References

- Docker Documentation — <https://docs.docker.com/>
- Docker Tutorial for Beginners — <https://www.youtube.com/watch?v=fqMOX6JJhGo>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.